TRANSLATORS

Agenda

- Translators
- Assemblers
- Compilers
- Interpreters
- References

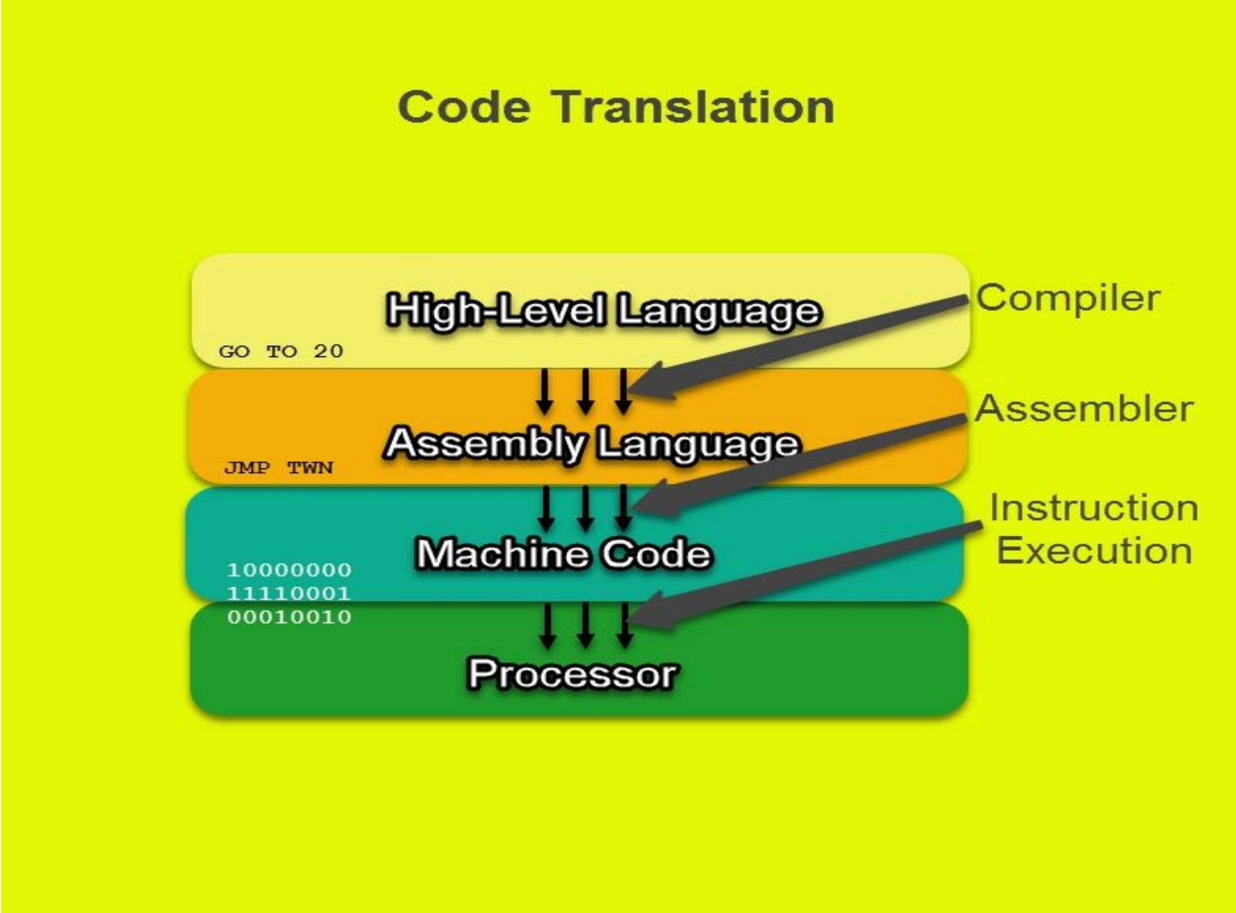
Translators (Language Processors)

- A computer understands instructions in machine code, i.e. in the form of 0s and 1s.
- It is a tedious task to write a computer program directly in machine code.
- The programs are written mostly in high level languages like Java, C++, Python etc. and are called source code.
- These source code cannot be executed directly by the computer and must be converted into machine language to be executed.
- Hence, a special translator system software is used to translate the program written in high-level language into machine code is called Language Processor (translator) and the program after translated into machine code (object program / object code).

Translators (Language Processors)

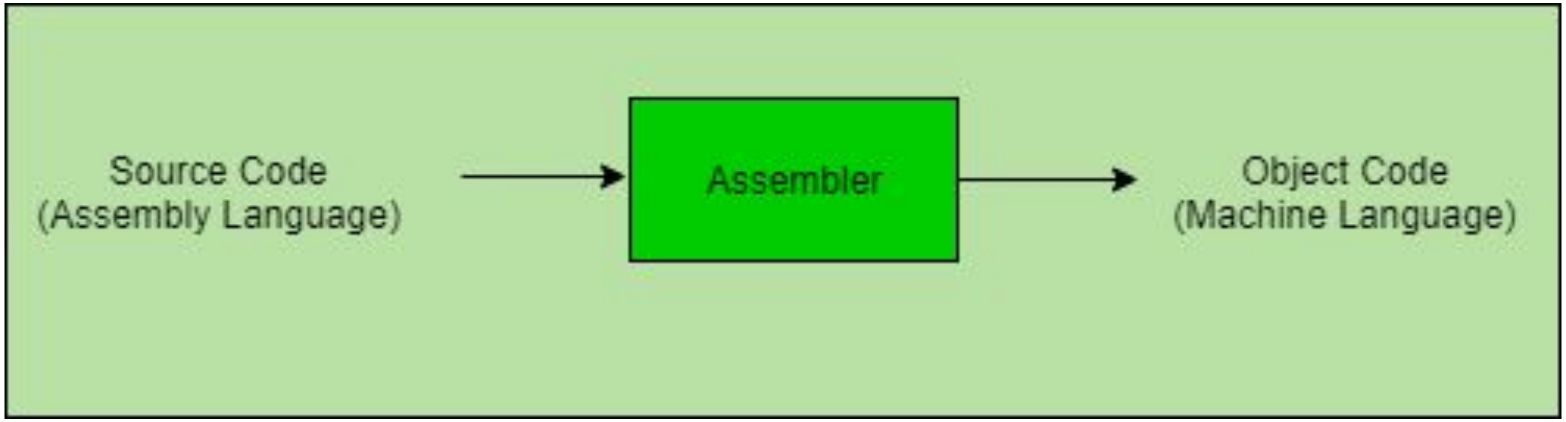
- Translator is a software code converting tool.
- A translator, could be a compiler, assembler, or interpreter.
- A compiler, assembler and interpreter translates a high-level programming language into machine code that the central processing unit (CPU) can understand.
- All translators, compilers, interpreters and assemblers are programs themselves.

Translators



Assemblers

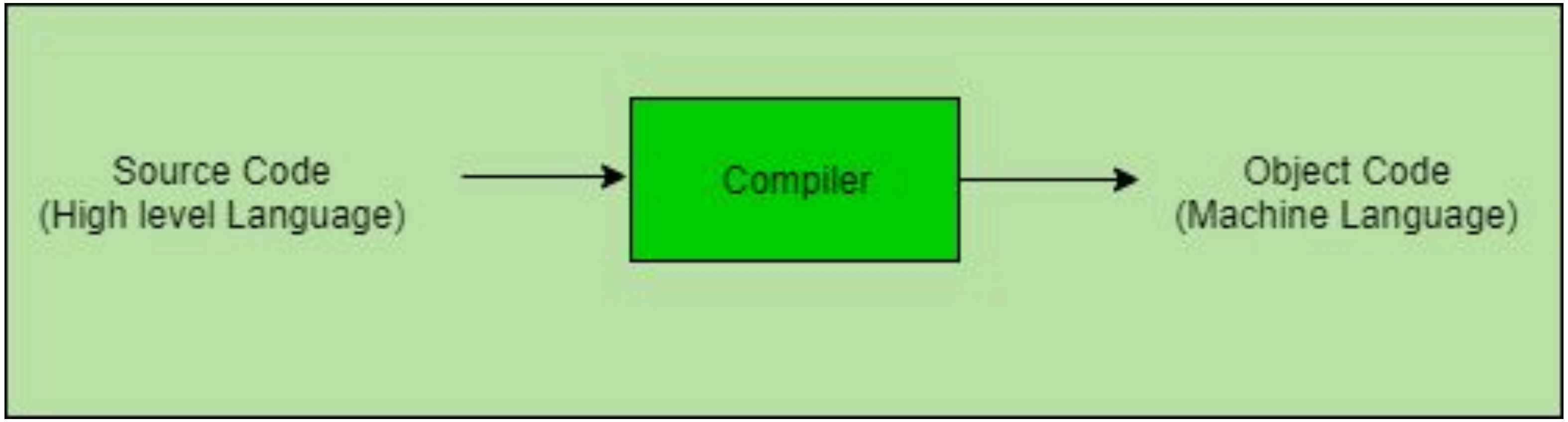
- An assembler translates a program written in assembly language into machine language.
- It is effectively a compiler for the assembly language, but can also be used interactively like an interpreter.
- Assembly language is a low-level programming language.
- An assembler converts assembly language code into machine code (also known as object code), an even lower-level language that the processor can directly understand.



Compilers

- The language processor that reads the complete source program written in high level language as a whole in one go and translates it into an equivalent program in machine language is called as a Compiler.
- In a compiler, the source code is translated to object code successfully if it is free of errors.
- The compiler specifies the errors at the end of compilation with line numbers when there are any errors in the source code.
- The errors must be removed before the compiler can successfully recompile the source code again.
- Compilers are also platform-dependent.
- Source-to-source compilers translate one program, or code, to another of a different language (e.g., from Java to C).

Compilers


Source Code
(High level Language)
Compiler
Object Code
(Machine Language)

Interpreters

- The translation of single statement of source program into machine code is done by language processor and executes it immediately before moving on to the next line is called an interpreter.
- If there is an error in the statement, the interpreter terminates its translating process at that statement and displays an error message.
- The interpreter moves on to the next line for execution only after removal of the error.
- An Interpreter directly executes instructions written in a programming or scripting language without previously converting them to an object code or machine code.
- An interpreter translates code like a compiler but reads the code and immediately executes on that code, and therefore is initially faster than a compiler.

Interpreters

- Thus, interpreters are often used in software development tools as debugging tools, as they can execute a single line of code at a time.

- Compilers translate code all at once and the processor then executes upon the machine language that the compiler produced.

- If changes are made to the code after compilation, the changed code will need to be compiled and added to the compiled code (or perhaps the entire program will need to be re-compiled.)

- Some examples of programming languages that use interpreters are Python, Ruby, Perl, Matlab and PHP.

THANK YOU